

Software Modeling and Design – Unit 2 Object Oriented Analysis (OOA)

1. Actor Identification, Actor Classification and Actor Generalization

Actor Identification: Process of identifying external users or systems interacting with the system.

Example: Customer, Admin, Payment Gateway.

Actor Classification:

- Primary Actor – Initiates interaction.
- Secondary Actor – Supports system operations.
- External Actor – External system interaction.

Actor Generalization: One actor inherits behavior from another actor.

Example: User → Customer, Admin.

2. Generalization, Include and Extend Relationships in Use Case Diagram

Generalization: Inheritance relationship ("IS-A").

Example: Payment → Credit Card Payment.

<<include>>: Mandatory functionality reuse.

Example: Withdraw Cash → Authenticate User.

<<extend>>: Optional functionality.

Example: Print Receipt → Withdraw Cash.

3. Purpose of Use Case Diagram

- Captures functional requirements.
- Shows interaction between actors and system.
- Defines system scope.
- Improves communication among stakeholders.
- Helps in system design and testing.

4. Qualified Association and Generalization

Qualified Association: Association with qualifier/key.

Example: Customer — [Account No] — Account.

Generalization: Inheritance relationship.

Example: Employee → Manager.

5. Qualified Association, Aggregation and Composition

Qualified Association: Uses qualifier for identifying object.

Example: Library — [Book ID] — Book.

Aggregation: Weak whole-part relationship.

Symbol: Hollow Diamond (◻).

Example: Department ◻ Employee.

Composition: Strong whole-part relationship.

Symbol: Filled Diamond (◼).

Example: House ◼ Room.

6. Difference Between Aggregation and Composition

Aggregation:

- Weak relationship
- Parts exist independently
- Hollow diamond (◻)

Composition:

- Strong relationship
- Parts depend on whole
- Filled diamond (◼)

7. Object and Class

Class: Blueprint/template of objects.

Example: Student class.

Object: Instance of class.

Example: s1 : Student.

8. Composition and Aggregation Relationships

Aggregation Example: Department ◻ Employee.

Employees exist independently.

Composition Example: House ◼ Room.

Rooms cannot exist without house.

9. ATM Machine Use Case Diagram (Important Points)

Actors:

- Customer
- Bank Server

Use Cases:

- Insert Card
- Authenticate User
- Withdraw Cash
- Deposit Money
- Check Balance

10. Medical Clinic Use Case Diagram

Actors:

- Patient
- Receptionist

Use Cases:

- Request Appointment
- Check Available Time Slot
- Schedule Appointment
- Update Appointment Book

11. Online Internet Banking Use Case Diagram

Actors:

- Customer
- Payment Gateway
- Admin

Use Cases:

- Login
- View Balance
- Transfer Funds
- Pay Bills
- Transaction History

12. Library Management System Use Case Diagram

Actors:

- Member
- Librarian

Use Cases:

- Search Book
- Issue Book
- Return Book
- Calculate Fine

13. E-Commerce Website Use Case Diagram

Actor:

- Customer

Use Cases:

- Search Item
- Add to Cart
- Calculate Cart Total
- Make Payment

Relationships:

- <<include>> → Calculate Total
- <<extend>> → Apply Coupon
- Generalization → Payment Methods

14. Company Class Diagram

Classes:

- Company
- Department
- Office
- Employee
- Manager

Relationships:

- Company ◆ Department
- Manager IS-A Employee
- Department located in Offices

15. College Library Management System Class Diagram

Classes:

- Library

- Book
- Member
- Librarian
- Transaction
- Fine

Relationships:

- Library ◆ Book
- Member — Transaction
- Transaction ■ Fine

16. Quick Revision Points

Class → Blueprint

Object → Instance

Aggregation ■ → Weak relationship

Composition ◆ → Strong relationship

Include → Mandatory

Extend → Optional

Generalization → IS-A relationship

Use Case Diagram → User interaction

Class Diagram → System structure